Plant Communications
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CellPress
Partner Journal

PCMD: A multilevel comparison database of intra- and cross-species metabolic profiling in 530 plant species

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https://doi.org/10.1016/j.xplc.2024.101038

ABSTRACT

Comparative metabolomics plays a crucial role in investigating gene function, exploring metabolite evolution, and accelerating crop genetic improvement. However, a systematic platform for intra- and cross-species comparison of metabolites is currently lacking. Here, we report the Plant Comparative Metabolome Database (PCMD; http://yanglab.hzau.edu.cn/PCMD), a multilevel comparison database based on predicted metabolic profiles of 530 plant species. The PCMD serves as a platform for comparing metabolite characteristics at various levels, including species, metabolites, pathways, and biological taxonomy. The database also provides a number of user-friendly online tools, such as species comparison, metabolite enrichment, and ID conversion, enabling users to perform comparisons and enrichment analyses of metabolites across different species. In addition, the PCMD establishes a unified system based on existing metabolite-related databases by standardizing metabolite numbering. The PCMD is the most species-rich comparative plant metabolomics database currently available, and a case study demonstrates its ability to provide new insights into plant metabolic diversity.

Key words: comparative metabolomics, plant database, multilevel comparison, metabolite characteristics

Hu Y., Ruan Y., Zhao X.-L., Jiang F., Liu D., Zhu Q., Zhang Q.-Y., and Yang Q.-Y. (2024). PCMD: A multilevel comparison database of intra- and cross-species metabolic profiling in 530 plant species. Plant Comm. **5**, 101038.

INTRODUCTION

Plant metabolites are essential for the growth and development of plants (Shen et al., 2023). Comparative metabolomics provides a strategy for understanding the complexity of plant metabolism. In recent years, comparative metabolomics has gained broad application in discerning the differential metabolites between various plant species and dissecting their nutritional value (Shi et al., 2022; Tang et al., 2022; Shen et al., 2023). Moreover, studying metabolomics across different species has provided invaluable insights into the evolution and domestication of crop metabolomes (Alseekh et al., 2021; Li et al., 2023a). Metabolomics analyses of crops such as maize, rice, and lettuce have successfully identified key metabolites linked to evolution and domestication (Deng et al., 2020; Zhang et al., 2020). In addition, integrating metabolomics with multiomics analyses provides substantial support for exploring the relationships among metabolites and complex agronomic traits (Cao et al., 2022; Li et al., 2023b). Overall, comparing metabolic profiles across various plant species holds significant promise for unraveling the composition of phytochemicals and aiding in the identification of species-specific accumulation of active metabolites.

Several databases have been constructed to investigate the roles of metabolites in plant growth and development, including Plant Reactome (Naithani et al., 2020), Plant Metabolic Network (PMN) (Hawkins et al., 2021), and RefMetaPlant (Shi et al., 2024). These databases serve as valuable resources for exploring plant metabolites from various perspectives. Plant Reactome, a knowledgebase and resource for comparative

Published by the Plant Communications Shanghai Editorial Office in association with Cell Press, an imprint of Elsevier Inc., on behalf of CSPB and CEMPS, CAS.

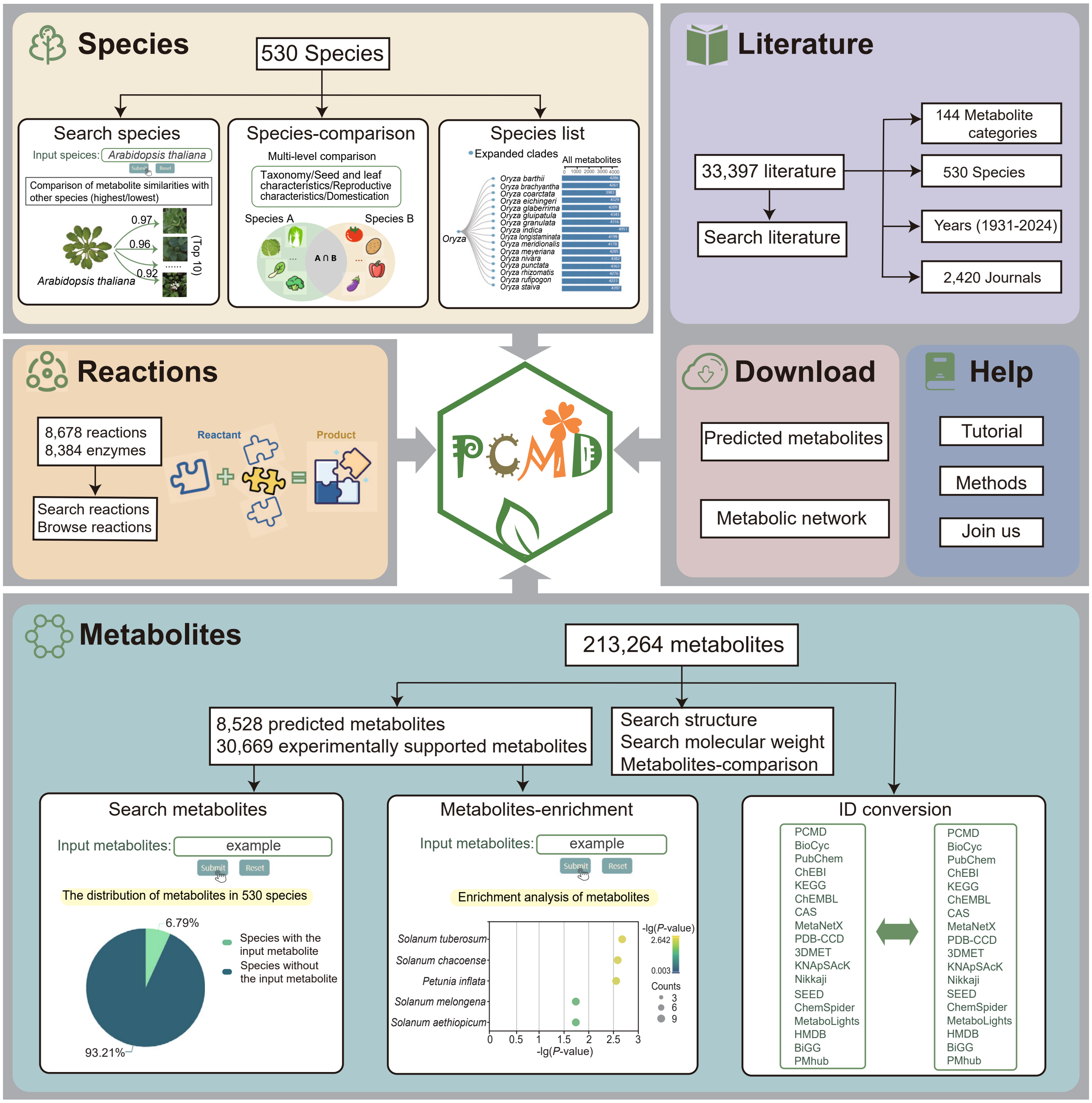

Figure 1. Overview of the PCMD.
Six modules are integrated into the PCMD: species, metabolites, reactions, literature, download, and help. Within each module, the principal datasets and representative functionalities are presented.

pathway analysis of 130 plant species, predicts reactions and metabolic pathways using *Oryza sativa* (rice) as a reference genome and provides an online tool for comparing pathways between rice and other plant species (Naithani et al., 2020). PMN focuses on metabolic networks for 126 plant species and includes functions for searching for relevant information on metabolic pathways and assisting in genome annotation (Hawkins et al., 2021). RefMetaPlant provides reference metabolome for 153 plant species, as well as several tools for analyzing plant metabolomes and metabolite identification (Shi et al., 2024). Despite these advances, there is a need for improved availability and connectivity of data between published databases, and a platform for comparative analyses of intra- and cross-species metabolites is still lacking.

In this study, we constructed the Plant Comparative Metabolome Database (PCMD), a multilevel comparison database for intra- and cross-species metabolic profiling (http://yanglab.hzau.edu.cn/PCMD) (Figure 1). We used a genome-scale metabolic model (GEM), which has proven effective for predicting the presence of

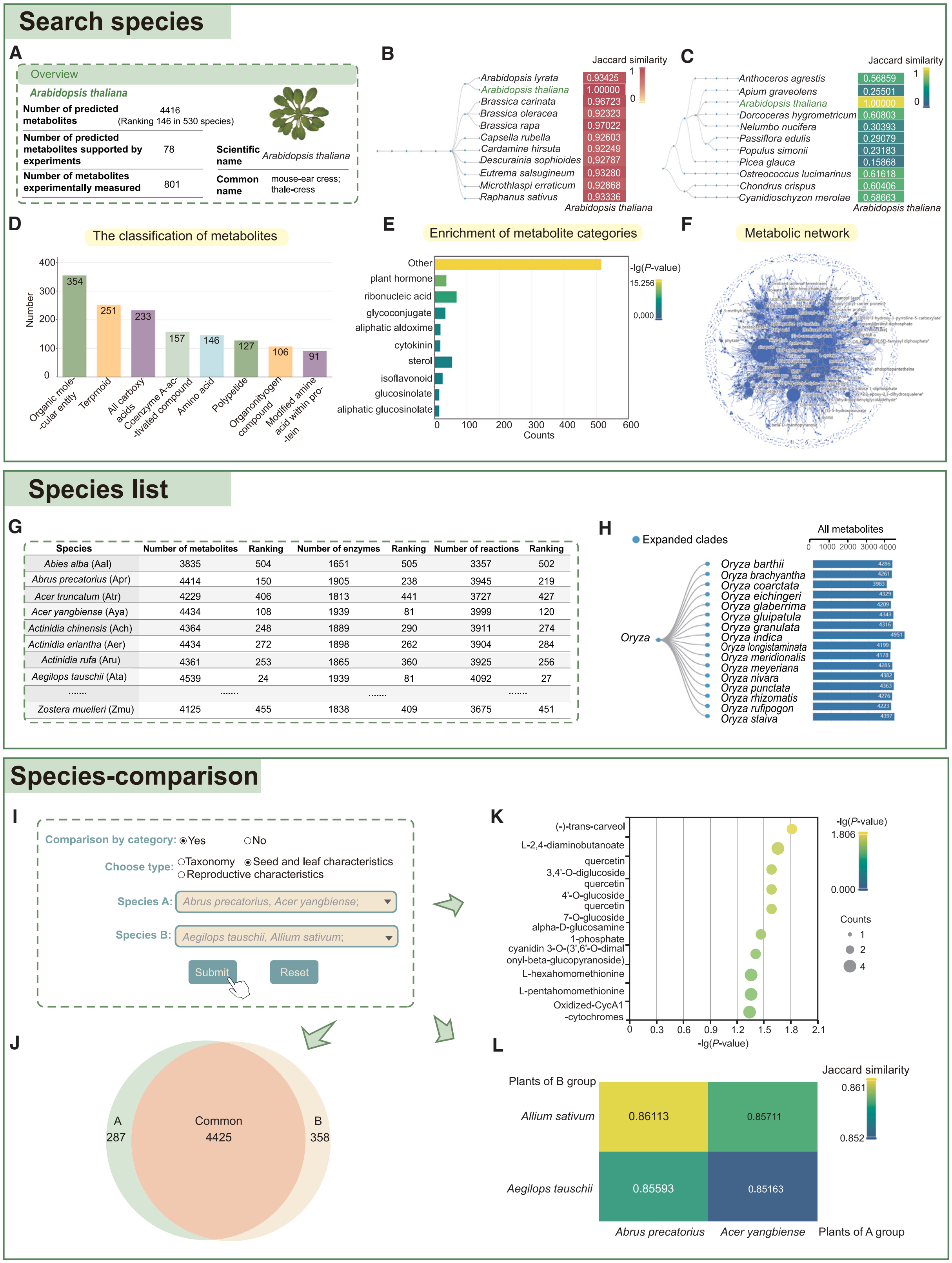

Figure 2. Search species, species list, and species comparison tool of the PCMD.
(A–F) Example of *A. thaliana*-related metabolite features on the search species page.
(A) Overview of metabolites in *A. thaliana*.
(B) Top 10 plants with the highest metabolite similarity to *A. thaliana*.

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metabolites in organisms on the basis of genomes (Mendoza et al., 2019), to predict metabolites, reactions, and metabolic networks for 530 plant species. The PCMD integrates abundant species classification information and provides various tools, including species comparison, metabolite enrichment, ID conversion, etc. These tools can be used to perform multilevel comparisons and enrichment analyses of metabolites across different species and to effectively convert metabolite IDs across multiple published metabolite-related databases. The PCMD will be a valuable resource for advancing comparative metabolomics research in plants.

RESULTS

Construction and overview of the PCMD

To understand the variations in metabolites among different plant species, genomic resources of 530 plant species were collected from published databases and recent plant genome assembly studies (Supplemental Table 1). In addition, enzymes and compounds from the KEGG and MetaCyc databases were obtained (Caspi et al., 2020; Kanehisa et al., 2021). Using the GEM (Thiele and Palsson, 2010) and the collected data, we predicted the metabolites and reactions of these 530 plant species and constructed the metabolic network of each plant by the RAVEN method (Wang et al., 2018) (Supplemental Figure 1A; see methods). Given the reliance on plant genome sequences, GEMs primarily predict potential primary metabolites inherent to each plant. To bridge this gap, 30 669 experimental metabolites were collected from MetaCyc and RefMetaPlant (Caspi et al., 2020; Shi et al., 2024). The collected information and the results of annotating predicted metabolites based on it were integrated into the PCMD. To facilitate comparisons of metabolite characteristics within and across species, extensive species classification information was collected (see methods), including taxonomy, reproductive characteristics, seed and leaf characteristics, and domestication information (Supplemental Figure 1B; Supplemental Table 2). Literature related to metabolites and species was also gathered, providing crucial references to support researchers in further exploration. In total, 213 264 metabolites, 8384 enzymes, 8678 reactions, 30 669 experimentally supported metabolites, and 33 397 literature references for 530 plant species were included (Supplemental Figure 1C). Finally, the PCMD (http://yanglab.hzau.edu.cn/PCMD) was constructed.

To meet the various needs of users for retrieval and analysis, the PCMD contains six distinct modules: species, metabolites, reactions, literature, download, and help (Figure 1). Within these modules, users can access abundant and convenient visual tools, including species comparison, metabolite enrichment, ID conversion, and others, to further compare metabolic differences, explore the enrichment of metabolites across multiple plant species, and perform ID conversion of metabolites across multiple databases (Figure 1).

Tools for multilevel comparison of metabolites across species

Unlike previous plant metabolite databases, the PCMD offers user-friendly online tools that enable multilevel comparisons of metabolic differences across species. Each species is detailed on a dedicated page, which provides valuable information on its metabolite characteristics. Through the search species page, users can easily access metabolic details by entering the scientific name, common name, or abbreviated name of a species. For example, the metabolic characteristics of *Arabidopsis thaliana* (*A. thaliana*) were examined, with tables showing predicted and experimentally measured metabolites as well as predicted protein–metabolite pairs for user convenience (Figure 2A). We then used the Jaccard similarity coefficient to compare metabolite similarity among 530 plant species in the PCMD (see methods). On each species page, users can explore plants with the highest and lowest metabolite similarity to the input species. Specifically, on the *A. thaliana* page, the results showed that *Brassica rapa* metabolites exhibited a relatively higher similarity to *A. thaliana* metabolites (with a Jaccard similarity coefficient of 0.970), and *Picea glauca* metabolites showed a relatively lower similarity (with a Jaccard similarity coefficient of 0.159) (Figures 2B and 2C), contrasting with the closer evolutionary relationship of *A. thaliana* to species such as *A. helleri*, *A. lyrata*, and *A. arenosa* in the Brassicaceae phylogenetic tree (Mandáková et al., 2017; Chen et al., 2022). This discrepancy could be attributed to the fact that the evolution of biosynthetic pathways is driven mainly by gene duplication (Lichman et al., 2020; Li et al., 2023a). Moreover, metabolome annotation may also be affected by the quality of the genome assembly, the accuracy of compound annotation information, the diversity of algorithm selection, and the completeness of sample collection. Therefore, it is necessary to integrate additional evidence such as genomic analysis, functional characterization of metabolites, and phylogenetic relationships to deepen our understanding of metabolite evolution (Li et al., 2023a). In addition, the PCMD categorizes predicted metabolites according to their uses (see methods), thus enabling the identification of the most enriched metabolites and metabolite categories within each species

(C) Top 10 plants with the lowest metabolite similarity to *A. thaliana*. **(D)** Top eight categories in *A. thaliana* metabolite classification.
(E) Top 10 metabolite categories with the highest enrichment levels in *A. thaliana*.
(F) Metabolic network of *A. thaliana*.
(G) Ranking of metabolites, enzymes, and reactions of 530 species displayed on the species list page.
(H) *Oryza* serves as an example of a species list page that displays species taxonomic relationships and metabolite distributions.
(I–L) Example of comparing metabolite differences between species A group (*Abrus precatorius* and *Acer yangbiense*) and species B group (*Aegilops tauschii* and *Allium sativum*) using the species comparison tool.
(I) Pipeline for comparison of plant metabolite differences between species A group (*Abrus precatorius* and *Acer yangbiense*) and species B group (*Aegilops tauschii* and *Allium sativum*) by choosing seed and leaf characteristics mode.
(J) Comparison of metabolite differences between species A group and species B group.
(K) Enrichment of common metabolites in species A group and species B group.
(L) Metabolite similarities between species A group and species B group.

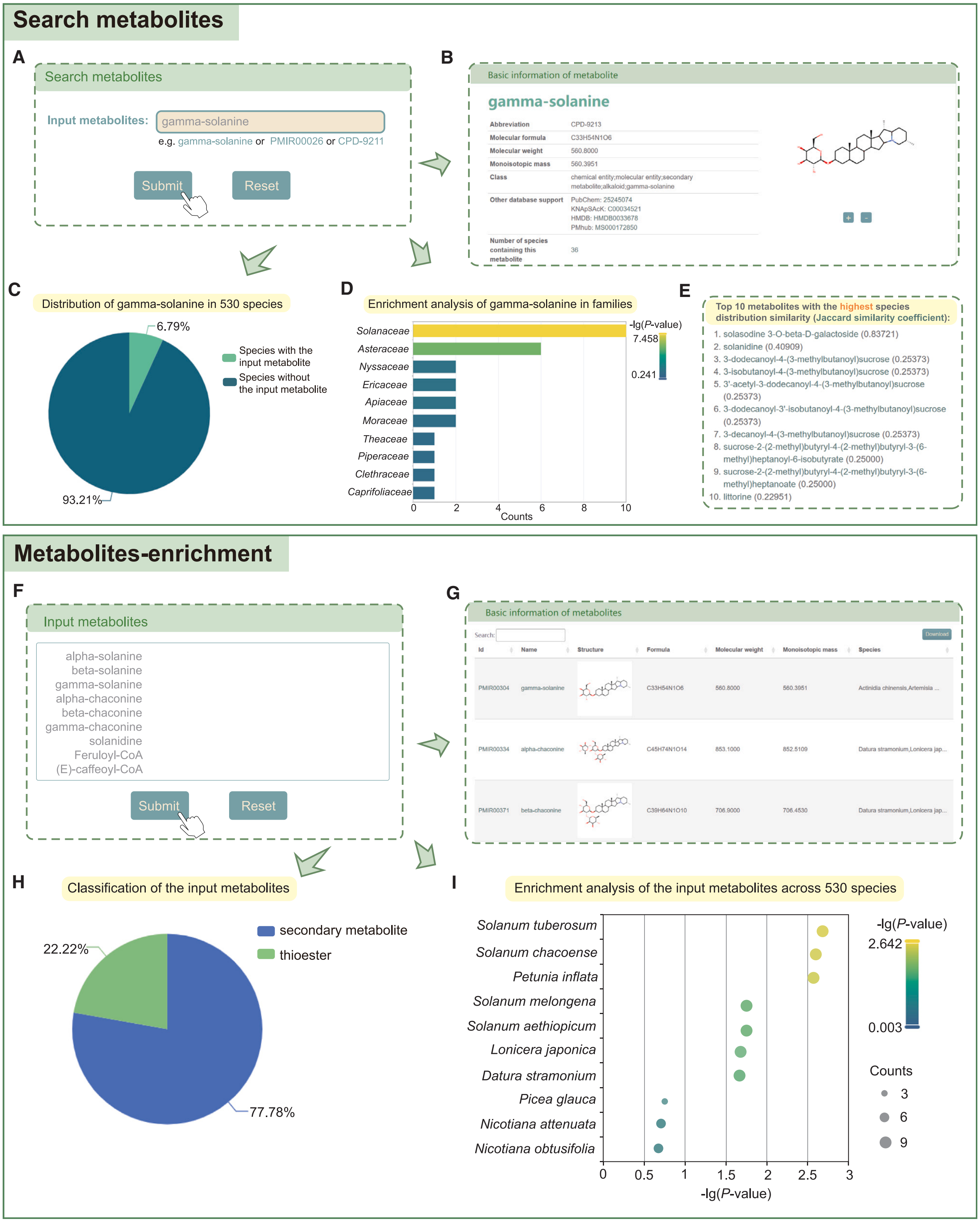

Figure 3. Search metabolites and metabolite enrichment tools of the PCMD.
(A–E) Example of gamma-solanine-related metabolic information on the search metabolites page.
(A) Pipeline for searching gamma-solanine on the search metabolites page.

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(Figures 2D and 2E). Users can also query the predicted metabolic network and related metabolites of the species (Figure 2F).

To compare differences in metabolites among various species, the PCMD ranks 530 plant species on the basis of the number of metabolites, enzymes, and reactions (Figure 2G). Users have the flexibility to search species using rankings on the species list page and explore the distribution of predicted metabolites across different plants (Figure 2H). In addition, the species comparison tool enables users to compare the specificity and commonness of metabolites between two different groups of plants at multiple taxonomic levels. When users choose a comparison by category, the PCMD provides three perspectives for comparison: taxonomy, seed and leaf characteristics, and reproductive characteristics. For example, when selecting the example plants provided in seed and leaf characteristics (Figure 2I), users can obtain results including a comparison of metabolite differences (Figure 2J), an enrichment analysis of differential metabolites (Figure 2K), a comparison of metabolite similarity (Figure 2L) between two groups of plants, etc. Alternatively, users can select "no" in the comparison by category option and upload two sets of plants to compare metabolite differences as needed (Figure 2I). In summary, the PCMD effectively facilitates multilevel comparisons of metabolites across various species, offering a valuable resource for researchers.

Services for metabolite enrichment analysis

To investigate the characteristics of metabolites in various species, the PCMD provides diverse tools such as search metabolites, search reactions, and metabolite enrichment. The search metabolites page provides users with convenient access to metabolite-related information, allowing for a deeper understanding of metabolite distribution and function. For example, users can search for "gamma-solanine" and access its basic information, distribution in 530 plant species, and families in which it is most abundant (Figures 3A–3D). The PCMD also provides distribution similarities between gamma-solanine and other metabolites among species, facilitating the discovery of similar metabolic pathways and functional characteristics across different species (Figure 3E). The search reaction page allows users to explore the involvement of metabolites in important biological functional reactions (Supplemental Figure 2A), providing detailed information about the reactions and the pathways in which they participate (Supplemental Figures 2B and 2C). Notably, if users wish to explore in which species a set of metabolites is enriched, the metabolite enrichment tool can achieve this purpose. For example, by uploading a set of metabolites (Figure 3F), users can obtain basic information on the metabolites (Figure 3G), their distribution in 530 species, classification and enrichment analysis of the metabolites (Figures 3H and 3I), and relevant literature. In addition, the metabolites comparison tool enables users to upload and annotate their own metabolite data by comparing them with the data in the PCMD. In summary, these tools greatly aid in investigating the enrichment characteristics of metabolites across various species, providing valuable insights for understanding biological diversity.

A standardized platform for integrating metabolites from various resources

Currently available plant metabolite-related databases offer a wide range of data resources for the investigation of metabolites from multiple perspectives (Naithani et al., 2020; Hawkins et al., 2021; Shi et al., 2024). However, differences in metabolite IDs across platforms indirectly impede the efficient integration and utilization of these resources. Therefore, it is essential to establish a standardized encoding for metabolite IDs across databases and provide a convenient ID conversion tool to facilitate their efficient utilization. In the PCMD, we have implemented a unified numbering system for all predicted and experimentally detected metabolites. In addition, we have established connections between these metabolites and existing metabolic databases, including BioCyc, PubChem, CHEBI, KEGG, CAS, ChEMBL, MetaNetX, Nikkaji, KNApSAcK, 3DMET, ChemSpider, SEED, PDB-CCD, HMDB, BiGG, MetaboLights, and PMhub. We then provide an ID conversion tool that enables users to efficiently convert metabolite IDs across these databases by inputting or uploading the metabolite IDs (Figure 4A). Our platform aims to effectively integrate metabolites from multiple databases and empower users to further understand the biological functions of metabolites. Moreover, we have annotated whether predicted metabolites in the PCMD have experimental validation, and users can quickly access supporting information from experimental sources and other databases (Figure 4B).

Some pages of the PCMD also provide links to other databases, such as MetaCyc, ChEBI, PubChem, and the National Center for Biotechnology Information (NCBI), enabling users to access additional information and resources. For instance, on the search species page, users can click on the species name to be directed to the NCBI website, where they can explore more information about the species (Figure 4C). Using the species comparison tool, users can navigate to MetaCyc for more detailed information on metabolic pathways associated with common or specific metabolites in the two groups of plants being compared (Figure 4D). In addition, literature references related to species or metabolites are included at the bottom of several pages in the PCMD, allowing users to explore

(B) Basic information about gamma-solanine.
(C) Distribution of gamma-solanine in 530 species.
(D) Enrichment analysis of gamma-solanine in the top 10 families.
(E) Top 10 metabolites with the highest species distribution similarity.
(F–I) Enrichment analysis of a group of metabolites using the metabolites enrichment tool.
(F) Pipeline for inputting a group of metabolites in the metabolites enrichment tool.
(G) Basic information on the input metabolites.
(H) Classification of the input metabolites.
(I) Enrichment results of input metabolites in the top 10 enriched species.

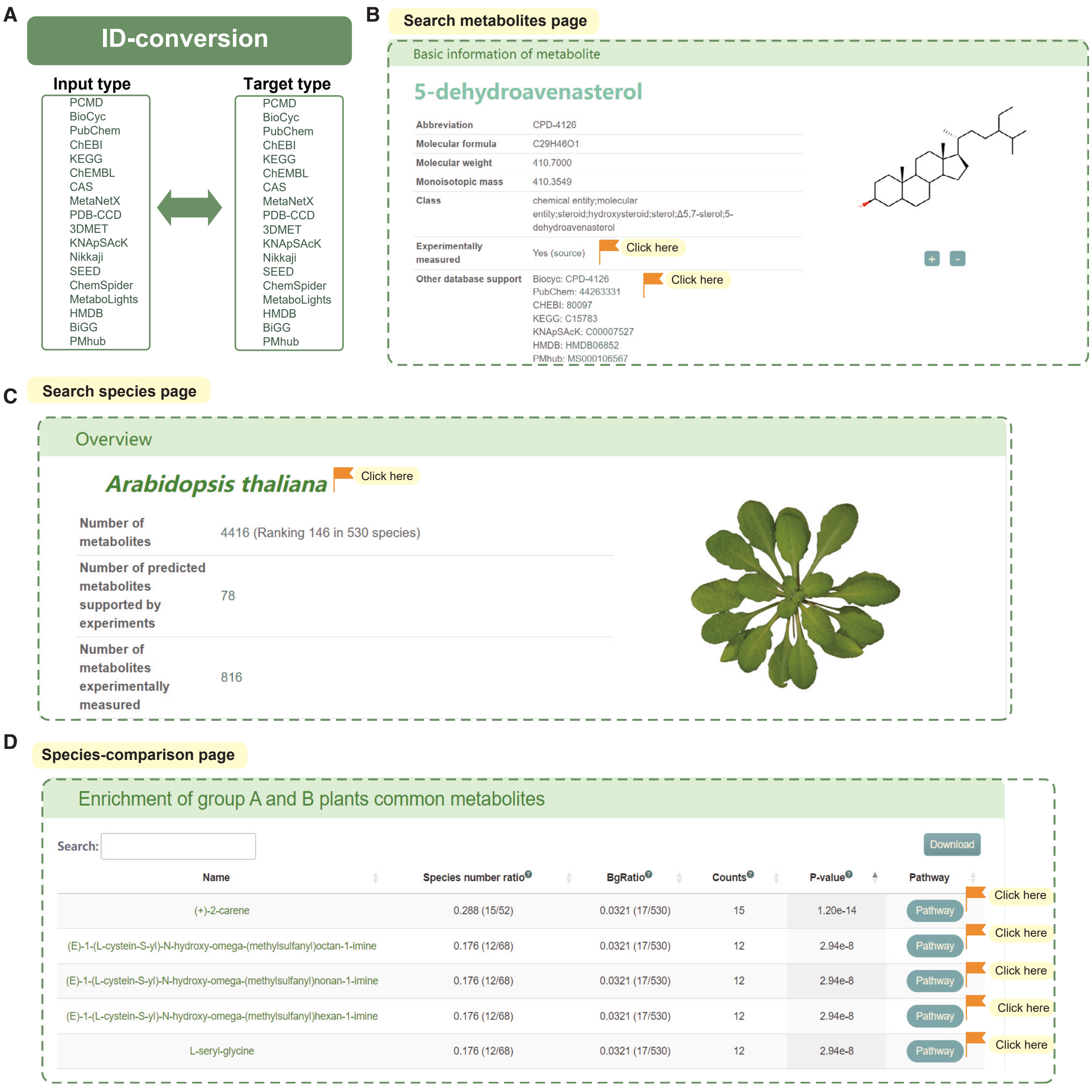

Figure 4. Efficient utilization of metabolite resources across multiple platforms.
(A) ID conversion tool in the PCMD.
(B–D) Examples of integrated metabolite analysis results from multiple platforms on the search metabolites page **(B)**, search species page **(C)**, and species comparison page **(D)**. The orange flag symbol indicates a location with external connections.

further by jumping to the NCBI website. The PCMD also features a literature module in which users can easily access relevant references by searching for metabolite categories, metabolites, species, publication years, or journals (Supplemental Figure 3). The PCMD assists users in accessing more pertinent research by filtering out duplicate and redundant literature, thereby saving time and enabling users to quickly comprehend relevant studies. Overall, the PCMD serves as an invaluable resource for advancing metabolomics research and promoting collaboration and knowledge exchange within the scientific community.

Case study: Comparison of metabolic differences between different genera

To demonstrate the capability of the species comparison tool for comparing metabolite differences across various levels, we performed a systematic analysis of metabolite variations between the genera *Brassica* and *Solanum*. The results revealed a total of 249 unique metabolites in *Brassica*, 320 unique metabolites in *Solanum*, and 4636 shared metabolites between the two genera (Figure 5A). Furthermore, we observed a metabolic similarity range of 0.741–0.870 between *Brassica* and *Solanum* species (Figure 5B), providing valuable insights into their metabolic

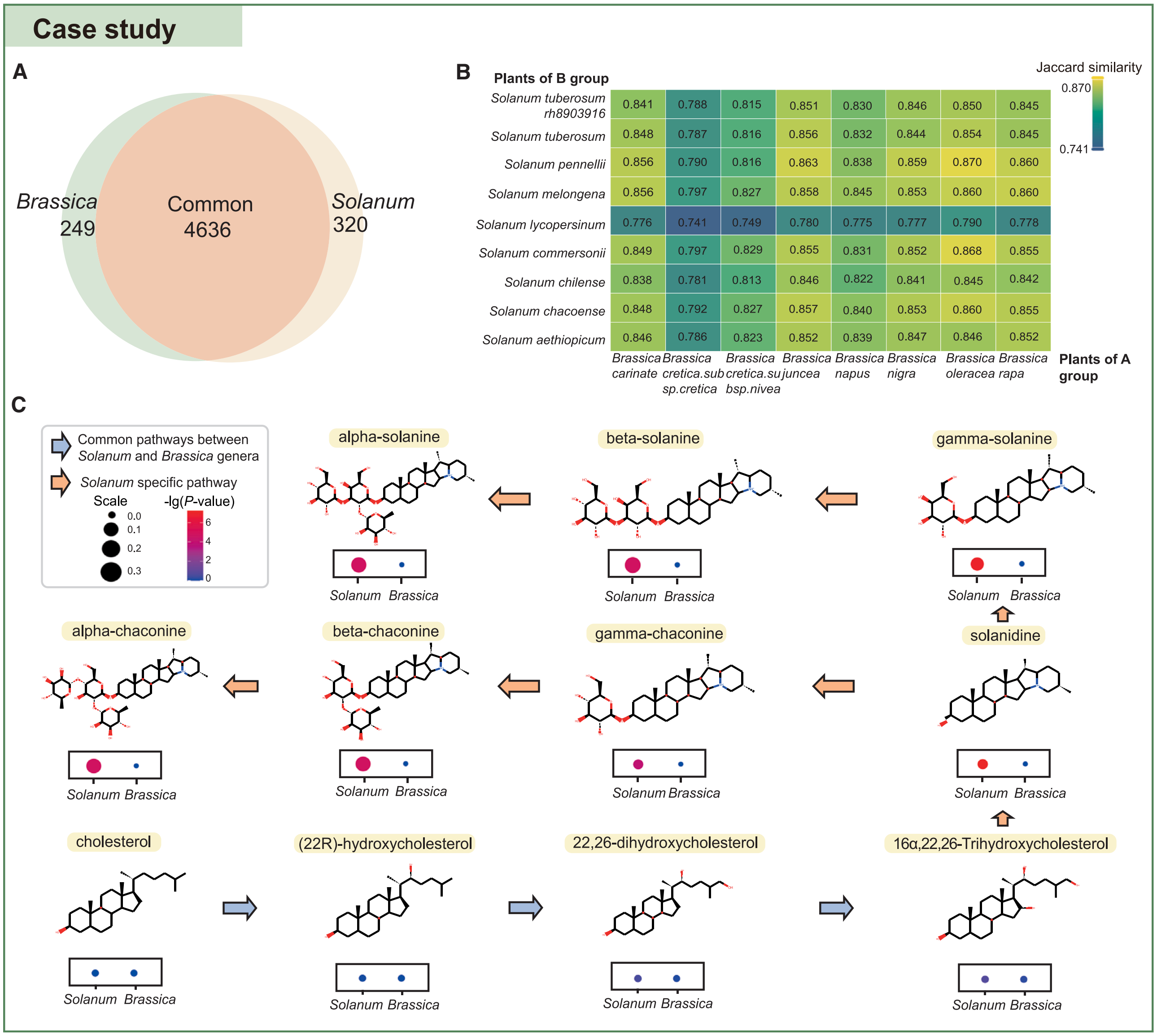

Figure 5. Case study on the comparison of metabolite differences between *Brassica* and *Solanum* species.
(A) Comparison of metabolite differences between the genera *Solanum* and *Brassica*.
(B) Metabolite similarities between *Solanum* and *Brassica*.
(C) Enrichment analysis of metabolites between *Solanum* and *Brassica*.

differences and commonalities. Notably, the comparative analysis identified alpha-chaconine and alpha-solanine as specific metabolites of the *Solanum* genus (Supplemental Table 3). These steroidal glycoalkaloids have been shown to play crucial roles in defense against insects and fungi in *Solanum* plants (Kuo et al., 2012). We also found that many metabolites involved in the upstream synthesis pathways of alpha-chaconine and alpha-solanine are present in both genera (Supplemental Table 3). For instance, a series of metabolites involved in the modification of cholesterol to 16α,22,26-trihydroxycholesterol and its further conversion to solanidine are included (Sawai et al., 2014; Nakayasu et al., 2021). Next, to investigate the potential reasons for the specific presence of alpha-chaconine and alpha-solanine in *Solanum* plants, we performed enrichment analysis of their metabolites. The results showed that *Brassica* plants lack the pathway for conversion of 16,22,26-trihydroxycholesterol to solanidine, leading to impaired synthesis of alpha-chaconine and alpha-solanine (Figure 5C; Supplemental Table 4). By examining the metabolic pathways of *Solanum* and *Brassica* plants, we obtained a better understanding of the biochemical differences between the two genera, providing a theoretical basis for further exploration of their insect and fungal resistance properties. Therefore, researchers can use the species comparison tool to identify key metabolic differences among species and to make specific functional improvements in plants.

DISCUSSION

GEMs offer significant advantages for predicting the presence of metabolites in organisms (Mendoza et al., 2019). A recent study has shown that the GEM prediction method provides extensive metabolic information and robust functionality (Mendoza et al., 2019). In the realm of reconstructing GEMs for plants, the RAVEN tool (Wang et al., 2018), which leverages genomic data, represents a paradigm shift with notable merits. This automated assembly of metabolic networks integrates pivotal biochemical insights from databases such as KEGG (Kanehisa et al., 2021) and MetaCyc (Caspi et al., 2020), expanding the annotation range of metabolic networks in plants (Bernstein et al., 2021) and enhancing the ability to predict metabolite presence. By contrast, traditional methods for predicting plant metabolites involve comparison of an organism's genome with known pathways and enzymes from reference databases and identification of plant metabolic networks (Hawkins et al., 2021). However, this approach may lead to errors in gene annotation, with many genes categorized as hypothetical sequences of unknown function (Lobb et al., 2020). In addition, compared with the challenges of large-scale experimental determination of metabolites, the PCMD provides researchers with predicted metabolite data across a wide range of plant species. It serves as a valuable resource for generating hypotheses, guiding experimental design, and facilitating comparative analyses.

However, the GEM method also has certain limitations. The quality of the input genome assembly and the accuracy of known metabolic pathways are crucial for ensuring the reliability of the prediction results. The potential lack of specific information on unique plant species or specialized metabolic pathways in existing databases could lead to gaps in the reconstructed networks. Moreover, when it comes to predicting secondary metabolites, the GEM method may have shortcomings owing to the inherent complexity and diversity of plant metabolic processes. The biosynthesis of secondary metabolites involves specific enzymes and reactions and is significantly influenced by environmental factors such as light, temperature, and pressure, thus posing a challenge to the accurate simulation and prediction of GEMs. Future developments in the field should focus on addressing these challenges, potentially leading to more robust models capable of better investigating the metabolic intricacies of diverse plant species.

Metabolite heterogeneity presents challenges to the accurate measurement and prediction of metabolites. Experimental measurements of metabolites may be influenced by diverse factors, such as environmental conditions, developmental stages, and genetic differences among individuals or populations, thus leading to biases. For example, a study of 29 common soybean varieties identified 169 metabolites, 104 of which exhibited significant variation across the varieties (Lin et al., 2014). Root metabolite profiles of 19 *A. thaliana* accessions revealed that the variability between individual plants exceeded the natural variation between accessions and experimental batches (Mönchgesang et al., 2016). These findings highlight the complexity and dynamic nature of metabolic processes. Moreover, it is vital to recognize that although predictive models have the potential to anticipate both conserved and new metabolic pathways, there may still be discrepancies between predicted and experimental groups. Therefore, the combination of experimental data and predictive models is crucial for understanding metabolic processes. To facilitate a broader exploration of metabolites, the PCMD not only includes predicted metabolites but also collects experimentally determined metabolites supported by MetaCyc and RefMetaPlant (Caspi et al., 2020; Shi et al., 2024), helping researchers to further understand the metabolic characteristics of various plants.

The PCMD stands out from other published metabolite-related databases such as KEGG, PMN, Plant Reactome, and RefMetaPlant in several respects: plant and metabolite-related quantities, multilevel comparison and enrichment analysis of metabolites, and attention to the combination of metabolite IDs from published databases (Supplemental Tables 5–7). Specifically, (i) the PCMD is the most species-rich plant metabolomics database currently available. Users can easily search for species names, common names, and abbreviations on the search species page to quickly access information related to the metabolite characteristics of the input species. (ii) The PCMD is the first platform that enables the comparison of metabolic differences across various biological classification levels among different plants. Although Plant Reactome offers a similar function for comparing metabolic pathways of species, it uses rice as a reference pathway for comparison with the predicted pathways of other species (Naithani et al., 2020). (iii) The metabolite enrichment tool provided in the PCMD focuses on investigating the presence and enrichment of metabolites in different species, whereas PMN enables enrichment analysis of a group of genes or compounds of interest in specific pathways (Hawkins et al., 2021). (iv) The ID conversion tool designed in the PCMD establishes connections between metabolic data in 17 published databases, enabling users to integrate and utilize different data resources (Supplemental Table 7). This sets the PCMD apart from KEGG, which mainly converts gene or protein entry identifiers from outside databases to its own identifiers. In general, these databases offer a wealth of metabolite information and analysis tools from various perspectives, providing users with the flexibility to select the most suitable database according to their specific research requirements for effective querying and utilization.

In the future, we will focus on eudicots, particularly plants of the *Brassica* genus, and examine the diversity and distribution of flavonoid metabolites based on their essential roles in mediating the responses of plants to biological and non-biological environmental factors. Second, we plan to optimize the upload module to enable users to upload their own experimental data for comparative metabolite enrichment analysis. Finally, we will be introducing new tools that facilitate exploration and analysis, including the development of a metabolic network visualization tool to enable users to investigate the complex relationships between metabolites and genes.

In summary, the PCMD offers an invaluable database for future research on the comparison of intra- and cross-species metabolites, providing new insights into plant metabolic diversity. In addition, the contribution of PCMD extends to promoting the standardization of plant metabolite identification and exploring the functional similarities of metabolites across different species. Our continued efforts in these areas will further enhance the utility and relevance of the PCMD for researchers in the field.

METHODS

Collection of plant genomes and classification

The genomic resources for the 530 plant species in the PCMD were gathered from a variety of reliable sources, including the Ensembl Plants database (v.51), recent plant genome assembly studies, and other published plant databases (Supplemental Table 1). In addition, a wealth of species classification information was compiled, incorporating taxonomy, reproductive characteristics, seed and leaf characteristics, and domestication information. These classification data were sourced from the NCBI and a study conducted by Marks et al. (2021) (Supplemental Table 2).

Reconstruction of the plant metabolic network

Plant metabolites were predicted using GEMs, which are computational simulations that represent all metabolic reactions occurring within a cell (Thiele and Palsson, 2010). In this study, the metabolic network for each plant was reconstructed *de novo* using the automated modeling tool RAVEN 2.0 (Wang et al., 2018).

RAVEN uses the MetaCyc (Caspi et al., 2020) and KEGG databases (Kanehisa et al., 2021) in the process of *de novo* metabolic network reconstruction. The MetaCyc-based reconstruction module facilitates the identification of species-specific reactions and predicted metabolites by comparing amino acid sequences of enzymes to those in the MetaCyc database, generating a draft model. Similarly, the KEGG-based reconstruction module uses a hidden Markov model trained on genes annotated in KEGG to assess protein sequence similarities for the target species.

MetaCyc-based draft models were generated using the "getMetaCycModelForOrganism" function with default settings. KEGG-based draft models were generated using the "getKEGGModelForOrganism" function, which uses a hidden Markov model trained on eukaryotic sequences with 100% sequence identity to query the plant proteome. The resulting draft models were then integrated using the "combineMetaCycKEGGModels" function provided by RAVEN (Supplemental Table 8). This integration approach enabled us to leverage complementary information from both the MetaCyc and KEGG databases, thereby enhancing the integrity and accuracy of the final draft model. The above analysis was performed in MATLAB 2018a.

The complete metabolic networks were exported in Excel format, and the resulting combined model integrates both KEGG and MetaCyc identifiers for metabolites. To ensure consistency and accuracy, we carefully reviewed and corrected the metabolite and reaction identifiers within the model and addressed any potential absence of reaction formulas for certain reactions. The summary of predicted metabolites for the 530 species is provided in Supplemental Table 9.

Comparative analysis of metabolite similarity among species

The Jaccard similarity coefficient was used to assess the similarity between two sets, such as sets of metabolites in different species. It is calculated by dividing the number of elements common to both sets (the intersection) by the total number of distinct elements present in either set (the union). Given two sets of metabolites, set A and set B, the Jaccard similarity coefficient, denoted as J, is represented by the following equation:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

Here, $|A \cap B|$ represents the number of elements in the intersection of sets A and B and $|A \cup B|$ represents the number of elements in the union of sets A and B. A ratio closer to 1 indicates higher similarity between the two sets, whereas a ratio closer to 0 indicates lower similarity.

Metabolite enrichment analysis

To assess the significance of specific metabolite enrichment within plant categories such as families, genera, or two groups of plants, we first constructed a presence-absence matrix. In this matrix, each row represents a metabolite, and each column represents a plant. If a metabolite is present in a plant, then the corresponding cell is marked as 1; otherwise, it is marked as 0. This approach enables us to visualize the presence of metabolites in each plant. We then performed hypergeometric testing to determine the statistical significance of specific metabolite enrichment in different plant categories. This method calculates the probability of observing a given number of metabolites in a particular plant category, assessing whether these metabolites are significantly enriched in that category. Specifically, we used the SciPy statistical package (v.1.7.3) (Virtanen et al., 2020) in Python (v.3.7.3) to calculate P values based on the hypergeometric distribution formula.

$$P(X = k) = \frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}}$$

$P(X = k)$ represents the probability that exactly k plants, out of a chosen subset of n plants, contain a specific metabolite. Here, N is the total number of plants. M is the number of plants that have a specific metabolite, and n is the number of plants in a specific category, such as a family or genus.

Metabolite classification

To standardize compound classification information across different databases, we combined the classification standards of MetaCyc (Caspi et al., 2020) and ChEBI (Hastings et al., 2016). ChEBI, which is maintained by the European Bioinformatics Institute, focuses primarily on classifying "small" chemical compounds on the basis of their chemical structure and biological relevance. On the other hand, MetaCyc categorizes compounds on the basis of their chemical structure and biological function, with an emphasis on functional aspects. Our approach predominantly used the MetaCyc classification system, complemented by the ChEBI classification system, to leverage the strengths of both databases. This approach ensured the accuracy and reliability of data through MetaCyc's focus on experimentally validated metabolic pathways while simultaneously broadening the scope of the database through ChEBI's extensive coverage of compounds.

In light of the complexity of compound chemical structures and subsequent varied levels of classification, we standardized the database's classification to a six-tier system. This system encompasses chemical entities, molecular entities, biochemical molecules (e.g., esters, alcohols), fundamental biomolecules (e.g., lipids, organic acids), cellular functions and signaling molecules (e.g., glycerolipids, steroids), and the compounds themselves. This classification simplifies the database structure, enhancing user understanding and navigation while ensuring data consistency and accuracy. The six-tier classification system provides sufficient detail to describe the chemical and biological attributes of compounds without overwhelming complexity, ultimately facilitating user understanding and retrieval.

System architecture and software for database construction

The PCMD (http://yanglab.hzau.edu.cn/PCMD) has been constructed using ThinkPHP (v.5.0.24) as the framework, with PHP serving as the foundational language. In addition, jQuery (v.3.6.0) is incorporated as the JavaScript library, and the Apache website server is used to parse the MySQL database. Data visualization is facilitated through the integration of various plugins, including Bootstrap (https://getbootstrap.com), DataTables (https://datatables.net), and Echarts (https://echarts.apache.org), enabling the creation of diverse charts such as bubble charts, pie charts, heatmaps, line charts, and network diagrams (Supplemental Table 8). The database is designed to provide a user-friendly experience and can be accessed online at no cost. It is also compatible with multiple browsers, including Chrome (recommended), Opera, Firefox, Microsoft Edge, and macOS Safari, ensuring accessibility across different platforms.

DATA AND CODE AVAILABILITY

The sources of all datasets are described in the supplemental information. All datasets are available at https://yanglab.hzau.edu.cn/PCMD/download.

SUPPLEMENTAL INFORMATION

Supplemental information is available at *Plant Communications Online*.

FUNDING

This research was supported by the National Natural Science Foundation of China (32322061 and 32070559), the Fundamental Research Funds for the Central University HZAU (2662022XXYJ007 and 2662023XXPY001) and the China Postdoctoral Science Foundation (2024M751038).

AUTHOR CONTRIBUTIONS

Q.-Y.Y. and Q.-Y.Z. conceived and supervised the study. Y.H., Y.R., X.-L.Z., and F.J. collected the datasets, performed the bioinformatics analysis, and constructed the database. Y.H., Y.R., and X.-L.Z. wrote the manuscript draft. Q.-Y.Y., Q.-Y.Z., Q.Z., and D.L. revised the manuscript.

ACKNOWLEDGMENTS

We gratefully acknowledge the invaluable contributions of Profs. ZhiNan Mei, Wei Chen, and Dexin Kong from the Huazhong Agricultural University of China, as well as the input of other researchers, for constructive feedback and suggestions for enhancing the PCMD. No conflict of interest is declared.

Received: May 16, 2024
Revised: June 19, 2024
Accepted: July 8, 2024
Published: July 11, 2024

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